



FCNSU-NET: Comprehensive recognition of the first trimester fetal CNS ultrasound standard planes and anatomical structures with linear transformer attention

Qingyu Zheng^a, Liya Li^a, Yubing Huang^a, Guorong Lyu^{a,*} , Pan Zeng^{b,*}

^a Department of Ultrasound, Second Affiliated Hospital of Fujian Medical University, Quanzhou, China

^b School of Medicine, Huaqiao University, Quanzhou, China

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ABSTRACT

Comprehensive recognition of first trimester fetal central nervous system ultrasound standard planes (FCNS-USP) and their anatomical structures is crucial for prenatal diagnosis, as accurate identification enables earlier detection of fetal brain abnormalities and timely intervention. However, this task remains challenging due to small structures, blurred boundaries, and unstable imaging.

This study proposes FCNSU-NET, a deep learning model designed specifically to achieve accurate identification of both FCNS-USPs and their key anatomical structures. To enhance anatomical structure detection, Agent Attention, SEN-CBAM, and CGRFPN are introduced into the YOLO detection framework, while a Plane Loss based on medical prior knowledge further improves detection accuracy and robustness. To further improve standard plane recognition, the detected anatomical structures are encoded by a GRU and introduced into the Swin Transformer, where they are fused with image features. This allows the classification module to jointly attend to both structural and image-level information, thereby enhancing recognition performance.

In this study, a total of 3,425 expert-annotated first trimester fetal CNS ultrasound images were collected, comprising 2,440 for training, 612 for testing, and 373 for independent clinical validation. On the test set, FCNSU-NET achieved an mAP@0.5 of 95.3% and mAP@0.5:0.95 of 67.1%, surpassing existing methods by 5.6 and 8.2 percentage points, respectively. The standard plane classification accuracy reached 98.53% with a specificity of 99.6%. On the clinical validation set, FCNSU-NET outperformed junior and intermediate physicians, and matched or exceeded senior experts (Kappa=0.90), demonstrating strong robustness and clinical applicability, demonstrating strong robustness and clinical applicability for automated prenatal ultrasound analysis.

1. Introduction

Obtaining the standard fetal central nervous system (FCNS) screening plane in the first trimester plays a crucial role in the screening and diagnosis of fetal CNS structural malformations. Artificial intelligence (AI) has recently become a valuable tool for the automatic identification and image quality control of fetal CNS ultrasound standard planes (FCNS-USP) during the second and third trimesters [1–4]. However, the application of AI in automatically identifying FCNS ultrasound screening planes during the first trimester has been rarely reported. In routine clinical practice, even when standardized ultrasound protocols are followed, the success rate and image quality of acquiring

standard planes often vary significantly due to fetal position constraints, the complexity of the intrauterine environment, and inter-operator variability in scanning techniques. This inconsistency in plane acquisition, resulting from both human and objective factors, substantially compromises the accuracy of fetal brain structure identification, the reproducibility of image interpretation, and the consistency of subsequent diagnoses. Therefore, promoting the standardized acquisition of ultrasound planes—particularly through the integration of artificial intelligence technologies to achieve more objective and automated image acquisition—has become a pressing challenge in the field of fetal neurosonography [5].

Fetal craniocerebral abnormalities are among the most common

* Corresponding authors.

E-mail addresses: lgr_feus@sina.com (G. Lyu), 954181775@qq.com (P. Zeng).

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congenital malformations, and ultrasound remains the primary method for diagnosing fetal CNS abnormalities, making it an essential component of prenatal screening. Despite this, most ultrasound examinations are performed during the second and third trimesters [6,7], even though many brain abnormalities originate in the first trimester [7]. Certain ultrasound signs observed in early pregnancy are considered potential markers for fetal anatomical abnormalities and may indicate latent risk. For example, when measuring the thickness of nuchal translucency (NT) in the fetal anterior-posterior midsagittal plane, three anechoic bands of approximately equal width above the occipital bone correspond to the brainstem, fourth ventricle, and cisterna magna, respectively [8,9].

Accurate acquisition of FCNS-USP in the first trimester is essential for ultrasound-based diagnosis. Previous studies have demonstrated [10] that using standard ultrasound planes in the first trimester yields a sensitivity of 85.7 %, specificity of 100 %, positive predictive value of 100 %, and negative predictive value of 99.9 % for detecting fetal CNS malformations. However, the identification of standard planes and key anatomical structures remains a challenging, labor-intensive, and operator-dependent task. AI models, particularly those based on deep learning, have shown promise in medical imaging and ultrasound standard plane recognition [11–15]. These models can achieve high recognition accuracy, reduce diagnostic errors across operators, and minimize dependence on physician experience. This is particularly beneficial for less experienced sonographers, as AI can serve as an auxiliary tool to accurately obtain standard planes and identify key anatomical structures, thereby improving diagnostic effectiveness [16,17].

In this study, we followed the guidelines for fetal ultrasound in the first trimester and recommendations from relevant expert panels [10–13]. Four standard FCNS-USPs were defined: Transverse through Lateral Ventricle (T-LV), Anterior-Posterior Midsagittal Plane (AP-MSP), Transverse through Thalamus (T-Th), and Coronal through Frontal Lobe (C-FL). The ultrasonic images of these four planes are shown in Fig. 1. To facilitate anatomical structure identification, ultrasound experts accurately annotated key anatomical features: Lateral Ventricle (LV), Choroid Plexus (P), Interhemispheric Fissure (IHF), Diencephalon (D), Brainstem (BS), Fourth Ventricle (4 V), Cisterna Magna (CM), NT, Thalamus (T), Tectum (Tec), Orbit Lobe (OL), and Nasal Bone (NB). LV, P, OL, and D refer to bilateral structures, and Tec denotes the complete mesencephalic tectal structure. The composition of each plane is illustrated in Fig. 1. After extensive data collection, a total of 3,425 images were gathered for this study, with 3,052 used for model development and 373 for clinical validation.

Despite recent advances, recognizing fetal CNS ultrasound images in the first trimester presents several challenges. The fetus at this stage is small, with incompletely developed anatomical structures, leading to small target regions, blurred boundaries, and low contrast. Frequent fetal movement and amniotic fluid effects contribute to unstable image

quality, noise, shadows, and artifacts. Additionally, dynamic CNS development in the first trimester results in significant anatomical variability across gestational weeks, making recognition even more difficult for AI models [18].

To address these challenges, this study proposes FCNSU-NET, a model designed for the comprehensive recognition of FCNS-USP and key anatomical structures in the first trimester. The contributions of this study are as follows:

(1) **A multi-task model for FCNS-USP recognition.** We collected 3,425 first-trimester FCNS ultrasound images, and based on this dataset, proposed FCNSU-NET. The model designed to simultaneously detect key anatomical structures and classify standard planes. Detected structures are encoded via GRU and fused with Swin Transformer [19] image features, enabling both accurate anatomical structure detection and robust plane recognition.

(2) **Multi-module and loss function design boost detection performance.** To improve detection of anatomical structures in FCNS-USP, we designed and introduced modules such as Agent Attention, SEN-CBAM, and CGRFPN, as well as a Plane Loss function that incorporates medical prior knowledge to guide the detection process and enhance prediction consistency. Experimental results demonstrate these improvements substantially increase feature representation and detection accuracy.

2. Related work

The rapid development of deep learning technologies has significantly advanced ultrasound imaging, improving image quality and processing efficiency—especially in real-time denoising and anatomical structure recognition. Cammarasana et al. [20] developed a deep learning framework for real-time denoising of ultrasound images, effectively preserving key anatomical features and facilitating high-quality, fast imaging suitable for real-time surgical guidance. Khan et al. [21] introduced a deep learning-driven beamforming technique, enhancing image quality by increasing the dynamic range by 15–20 dB, which is particularly valuable in complex ultrasound environments.

Deep learning has also enabled multi-task modeling in medical ultrasound. One of our previous studies [22] implemented a multi-task learning framework for standard thyroid ultrasound plane identification and anatomical structure localization, streamlining the workflow by eliminating the need for multiple models. Similarly, Ramirez Zegarra and Ghi [23] developed an automated fetal biometric analysis system using deep learning to reduce dependence on operator expertise and improve standardization and efficiency. Luijten [24] combined data-driven methods with domain-specific knowledge, enhancing ultrasound image quality and enabling effective learning from small datasets, thus facilitating the broader application of deep learning in ultrasound diagnostics.

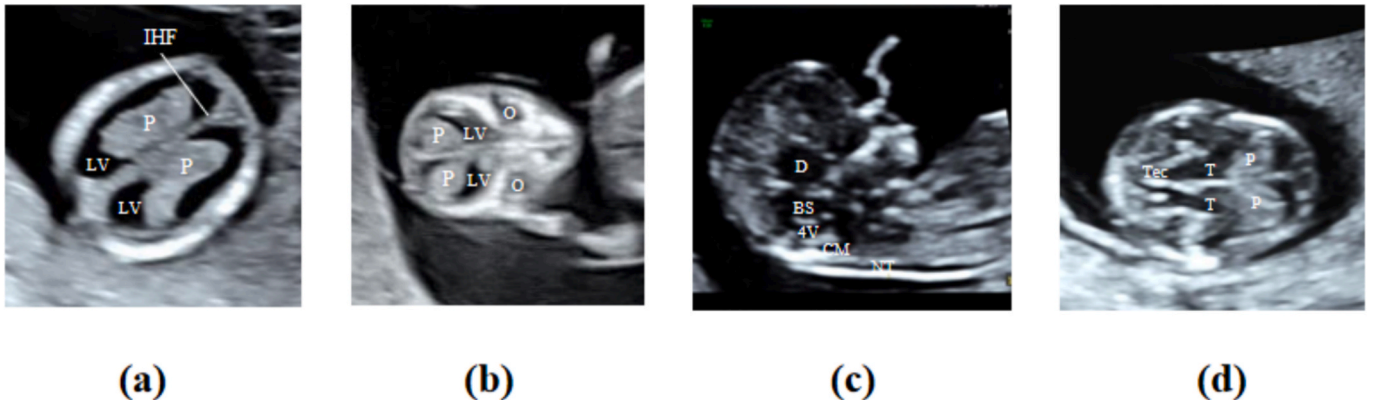


Fig. 1. Four standard planes of FCNS-USP. (a) T-LV, (b) C-FL, (c) AP-MSP, (d) T-Th.

For fetal ultrasound plane classification, self-supervised and ensemble learning approaches have been proposed. Feng et al. [25] developed a self-supervised learning algorithm that addresses data imbalance, significantly improving the speed and accuracy of fetal diagnosis. Krishna and Kokil [26] introduced a stacked deep learning ensemble for automatic detection of fetal standard planes, achieving 95.69 % accuracy in prenatal screening.

With these advancements, deep learning has made substantial contributions across ultrasound image denoising, beamforming, anatomical structure recognition, and automatic classification, significantly improving image clarity, diagnostic efficiency, and real-time performance. However, most existing works focus on later trimesters or general anatomical recognition, and their applicability to first-trimester fetal CNS standard plane recognition remains limited.

In the field of deep learning, the Transformer architecture [27] has garnered considerable attention for its robust feature extraction and long-range dependency modeling capabilities. In image analysis, these strengths allow for better capture of complex feature relationships and fine image details [28]. However, the Transformer model faces challenges due to the quadratic computational complexity of Softmax's self-attention mechanism, which grows rapidly with image resolution, significantly increasing memory consumption and processing time. These high computational demands limit the feasibility of real-time processing in practical applications.

To address these limitations, Liu et al. [29] proposed the Swin Transformer, which employs a hierarchical structure and local window-based self-attention, substantially reducing computational complexity and improving efficiency. Transformer-based models have been widely adopted in medical image analysis. For example, Chen et al. [30] proposed TransUNet, which integrates the multi-scale feature extraction of UNet with the global context modeling of Transformers, improving segmentation accuracy. Valanarasu et al. [31] introduced the Medical

Transformer, combining convolution and attention mechanisms to enhance key structure extraction and identification in medical images.

Despite these advances, significant challenges remain in the automatic recognition of fetal central nervous system ultrasound standard planes in the first trimester. Detecting fine anatomical structures is particularly difficult due to subtle boundaries and dynamic fetal development, while high model accuracy and real-time processing are critical for clinical application. To the best of our knowledge, there is still a lack of effective solutions capable of simultaneously recognizing both FCNS-USP and their key anatomical structures in the first trimester.

To overcome these challenges, this study proposes FCNSU-NET, an innovative deep learning model that leverages Transformer feature extraction, attention mechanisms, and medical prior knowledge for efficient and accurate recognition of first-trimester fetal CNS ultrasound images. FCNSU-NET is designed to provide a high-performance clinical application for anatomical structure detection and standard plane recognition in FCNS ultrasound images.

3. Method

3.1. Overview of the model framework

In this study, we propose a deep learning model, FCNSU-NET, for the recognition of FCNS-USP during the first trimester. The overall framework is illustrated in Fig. 2. The baseline model is enhanced by introducing Agent Attention, an advanced SEN-CBAM, CGRFPN, and a Plane Loss function, which guides the model to focus on key anatomical structures within each standard plane and improves detection precision.

The Swin Transformer is utilized for image feature extraction, while a GRU compresses and encodes the anatomical structures identified during the object detection phase, resulting in an anatomical information vector. This vector is then integrated into the Swin Transformer,

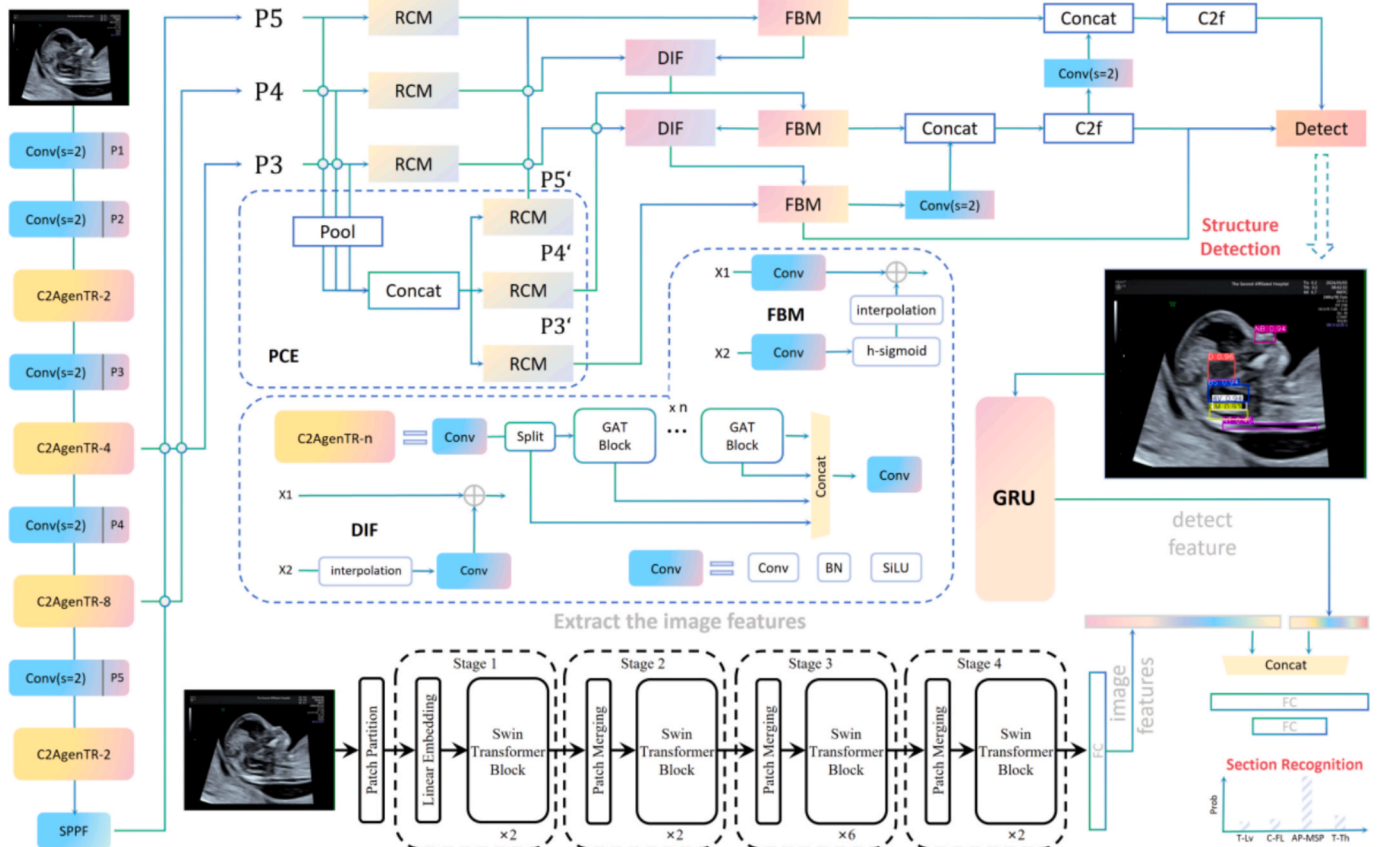


Fig. 2. Diagram illustrating the FCNSU-NET model framework.

facilitating more accurate identification of FCNS-USP in the first trimester. The following sections will detail the innovative aspects of our model.

3.2. Attention mechanism

Accurate identification of key anatomical structures and standard planes in first-trimester fetal central nervous system ultrasound images presents significant challenges, mainly due to the small size of early fetal structures, unclear boundaries, and fluctuating image quality. Transformer and attention mechanisms have demonstrated substantial advantages in modeling global contextual information, which facilitates better characterization of FCNS-USP features in medical ultrasound images. To further enhance the model's recognition of anatomical features and extraction of salient information, we introduce an attention mechanism in this study.

3.2.1. Agent attention

To address the trade-off between computational complexity and expressive power inherent in traditional Transformers for medical image processing, we introduce the Agent Attention mechanism proposed by Han et al. [32]. Agent Attention is a generalized linear attention mechanism that uses "agent tokens" as intermediaries between queries and key-value pairs. This approach achieves higher computational efficiency than traditional Transformer Softmax while retaining strong global context modeling capabilities.

The calculation process of Agent Attention is illustrated in Fig. 3. Agent Attention consists of two primary steps: Agent aggregation and Agent broadcasting. In the Agent aggregation step, agent tokens aggregate information from the key K and value V . The calculation formula for this step is as follows (where σ denotes the Softmax function):

$$V_A = \text{Attn}_s(A, K, V) = \sigma(AK^T)V \quad (1)$$

Next, the Agent Token AAA is used as the key to broadcast the aggregated information to the query Q , a process known as Agent broadcasting. The corresponding formula is:

$$O = \text{Attn}_s(Q, A, V_A) = \sigma(QA^T)V_A \quad (2)$$

Through these two steps, the final output of Agent Attention can be expressed as the following formula:

$$O = \sigma(QA^T)\sigma(AK^T)V \quad (3)$$

Agent Attention reduces the original computational complexity from $O(N^2)$ to $O(Nn)$, where n represents the number of proxy tokens, and $N \ll n$. This design not only maintains the global context modeling capability of the original Transformer, but also significantly reduces computational costs. As shown in Fig. 3, Agent Attention utilizes agent tokens to aggregate global information and distribute it into a single image tag, effectively integrating Softmax and linear attention.

3.2.2. Spatially enhanced convolutional Block attention module

The Convolutional Block Attention Module (CBAM) [33] is an effective attention mechanism that enhances the expressiveness of feature maps by incorporating both channel attention and spatial attention. While the original CBAM mechanism performed well across various visual tasks, the convolution operation in the spatial attention module did not fully exploit the features of each individual channel. To address this limitation, we propose an improved version of the CBAM attention mechanism, named SEN-CBAM (Spatially Enhanced Convolutional Block Attention Module), whose structure is shown in Fig. 4.

SEN-CBAM uses depth-separable convolution to compute spatial attention independently for each channel. In the improved SEN-CBAM, the calculation of spatial attention is realized by performing a channel-by-channel convolution operation for each channel. For the input feature graph $X \in \mathbb{R}^{B \times C \times H \times W}$, depth-separable convolution is used to generate spatial attention for each channel. SEN-CBAM is calculated as follows:

$$O_c = X_c \cdot \sigma(\text{PWConv}(\text{DWConv}(X)))_c \quad (4)$$

Firstly, depth-separable [34] convolution is applied to input feature maps x to generate feature maps for each channel. Next, point-by-point convolution (pw convolution) is applied to enhance the interactivity between channels. The spatial attention of each channel is calculated after obtaining the feature graph processed by point-by-point convolution. Finally, the feature map is multiplied with the corresponding spatial attention element by element to get the final output for each channel. With this design, SEN-CBAM provides a unique spatial feature weighting for each channel, enabling the model to more effectively capture the spatial importance of different channels in ultrasound images, improving the ability to focus on important structures and features.

3.2.3. Gated Agent Transformer Block

To balance the expressive power and computational efficiency of attention mechanisms, we designed a modular Gated Agent Transformer Block (GAT Block) that integrates Agent Attention and convolutional

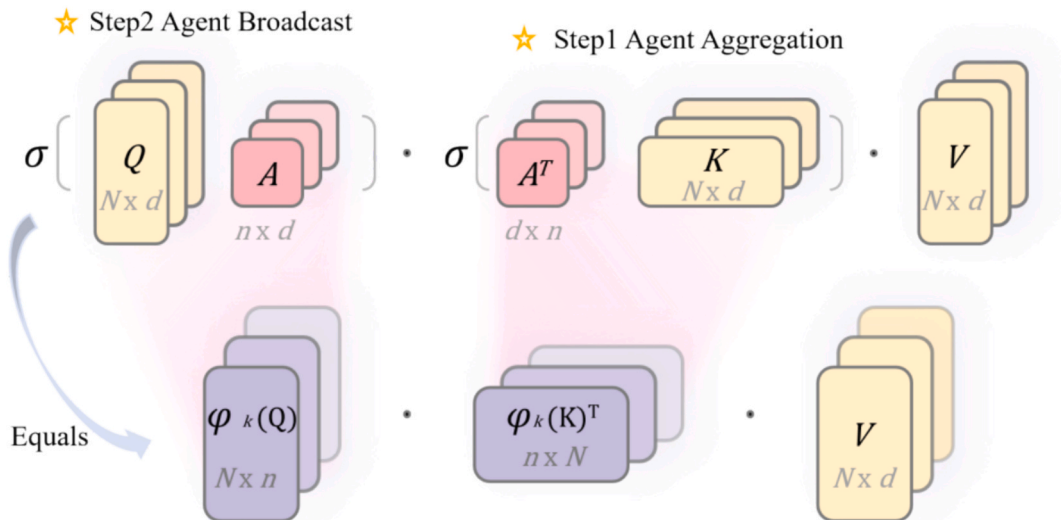


Fig. 3. Agent Attention uses agent tokens to aggregate global information and distribute it into a single image tag, enabling the actual integration of Softmax and linear attention.

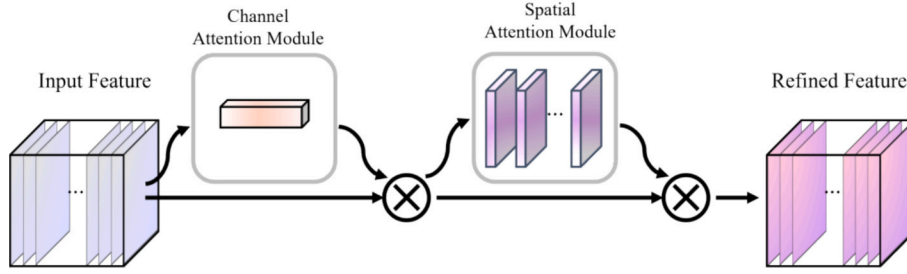


Fig. 4. The structure of SEn-CBAM.

attention mechanisms, as shown in Fig. 5. By combining the global feature extraction capability of Transformers with the local feature aggregation ability of convolutional neural networks, this module efficiently identifies key anatomical structures in ultrasound images.

In the GAT Block, Agent Attention captures the global context of each image part, allowing the model to focus effectively on important anatomical structures. To address the limitations of linear attention methods, we incorporate Depthwise Convolution (DWC) along with SEn-CBAM. Specifically, the DWC module enhances feature diversity, while SEn-CBAM calculates independent spatial attention to help the model accurately focus on key regions. This combination allows the model to capture global information while retaining fine-grained feature differences, improving its ability to model details. Such an approach effectively compensates for the limitations of single-method models, enhancing performance in dealing with small targets with complex backgrounds and blurred boundaries, thereby improving recognition accuracy and stability for FCNS ultrasound images in the first trimester.

The global and local features are then fused and integrated into the Gated Feed Forward Network (GFFN). The GFFN, equipped with a gating mechanism, allows the model to flexibly process features extracted by Agent Attention, DWC, and SEn-CBAM, thereby further optimizing feature representation.

3.3. Context-Guided spatial feature reconstruction feature pyramid network

The anatomical characteristics of the fetal central nervous system (FCNS) during the first trimester present several challenges for recognition due to their small size, fuzzy boundaries, and significant variation with gestational age. Ultrasound image quality during this period is often impacted by fetal position, amniotic fluid environment, and other factors, with noise and artifacts further complicating accurate recognition.

To address these challenges, we designed a Context-Guided Spatial Feature Reconstruction Pyramid Network (CGR-FPN) for effective target detection and feature extraction of FCNS ultrasound images. CGR-FPN integrates advanced feature extraction and fusion mechanisms to enhance recognition in complex backgrounds and across multi-scale targets in ultrasound.

The core modules of CGR-FPN include the Pyramid Context Extraction module (PCE), Dynamic Interpolation Fusion (DIF), Feature Fusion Module (FBM), and Rectangular Self-Calibration Module (RCM). The PCE module integrates multi-scale feature information to improve the model's context awareness and helps in better understanding multi-scale objects through hierarchical feature fusion. The DIF and FBM modules

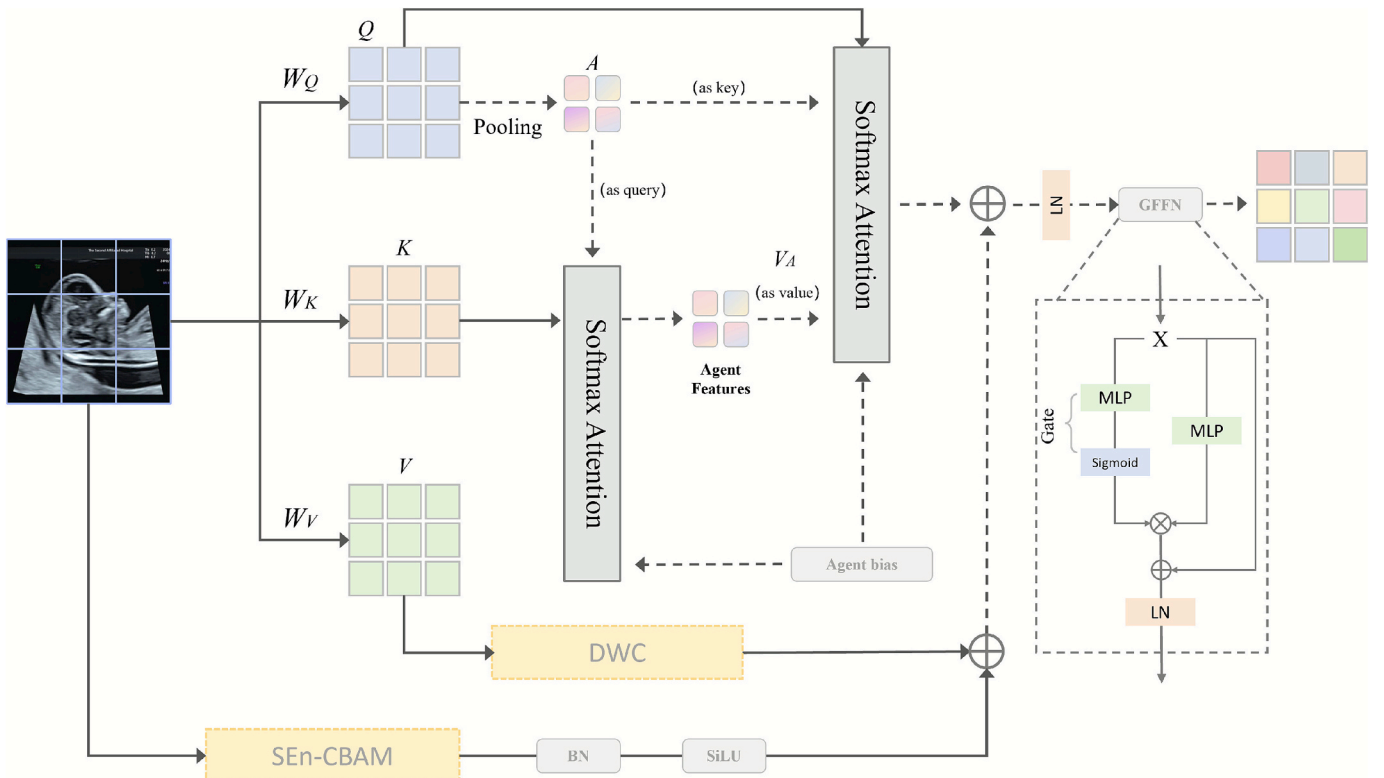


Fig. 5. GAT Block structure diagram.

enhance the fusion of multi-scale features via dynamic interpolation and feature-weighted fusion, enabling the model to better handle key features at different scales. The structures of the PCE, DIF, and FBM modules are shown in Fig. 2.

The structure of the RCM is presented in Fig. 6. The core component of the RCM module is the Rectangular Self-Calibrating Attention (RCA). RCA captures the global context in both horizontal and vertical directions and explicitly models the rectangular key regions through shape self-calibration, allowing the model to focus more effectively on key anatomical regions in the FCNS. This enhances the ability to reconstruct features and focus on important anatomical structures. The four steps of RCA feature extraction are detailed in the following formula:

$$\begin{cases} H_p(x) = \text{Pool}_h(x), & V_p(x) = \text{Pool}_v(x) \\ \tilde{y} = H_p(x) + V_p(x) \\ \xi_c(\tilde{y}) = \delta(\psi_{k \times 1}(\phi(\psi_{1 \times k}(\tilde{y})))) \\ \xi_F(x, y) = \psi_{3 \times 3}(x) \odot y \end{cases} \quad (5)$$

First, RCA is applied by horizontal pooling and vertical pooling to capture the axial global context, generating two independent axial feature vectors $H_p(x)$ and $V_p(x)$, by adding these two features together we get a rectangular area of attention $\tilde{y}$. The rectangular attention area is obtained by adding these two features together. Then, the attention diagram is calibrated using strip convolution of two large convolution kernels, which are adjusted independently in the horizontal and vertical directions, and the final attention weight graph $\xi_c(\tilde{y})$ is obtained. Where ψ stands for large convolution kernel strip convolution. k represents the size of the convolution kernel, ϕ represents the operation containing Batch Normalization and ReLU functions, and δ represents the Sigmoid activation function. In order to better fuse attention features with input features, 3x3 deep convolution is used in the module to further extract local details of input features, and the fused features are weighted by the Hadamard product to obtain the output $\xi_F(x, y)$ after RCA.

3.4. Gated Recurrent Unit

To enhance the feature extraction of key anatomical structures in FCNS-USP during the first trimester, we first utilized YOLO to accurately detect key structures within each plane. On this basis, to enable subsequent plane recognition models to effectively interpret these structures, we used Gated Recurrent Unit (GRU) [35] to extract features from the detected elements.

GRU regulates the flow of information through a gating mechanism, effectively mitigating the gradient vanishing problem common in traditional RNNs when dealing with long-term dependencies. The structure of GRU is illustrated in Fig. 7. Compared with LSTM [36], GRU has a simpler architecture, consisting of only an update gate and a reset gate, which makes its computation more efficient. The calculation process of GRU is shown in the following equation:

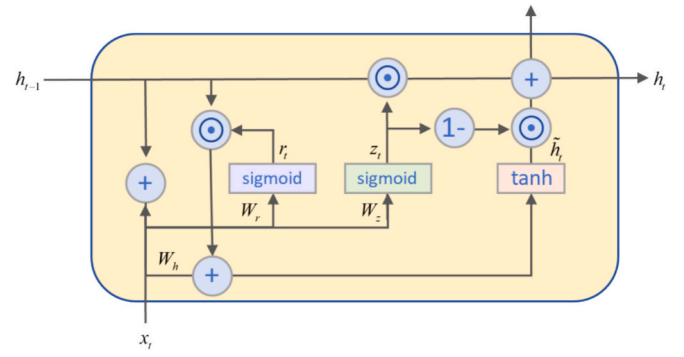


Fig. 7. GRU structure diagram.

$$\begin{cases} z_t = \sigma(W_z x_t + U_z h_{t-1}) \\ r_t = \sigma(W_r x_t + U_r h_{t-1}) \\ \tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1})) \\ h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \end{cases} \quad (6)$$

Where, x_t represents the input feature vector, h_{t-1} is the hidden state of the previous moment, z_t and r_t are the update and reset gates respectively, which are used to control information flow, $\tilde{h}_t$ is the candidate hidden state, and the final output h_t is the hidden state of the current moment.

In the plane recognition task, in order to continuous discrete feature information, we first map each target information into a 64-dimensional vector through a linear layer. Specifically, target features include category (cls), location information (x, y, w, h), and confidence (conf). This information is mapped through A linear layer to a 64-dimensional vector $t_i \in \mathbb{R}^{64}$. We then take these 64-dimensional feature vectors $\{t_1, t_2, [\text{HORELE}], t_T\}$ as input to the GRU, which processes a sequence of multiple detection target features and finally outputs a 256-dimensional vector $h_T \in \mathbb{R}^{256}$ that combines all the target features.

Finally, the GRU output h_T is concatenated with the 768-dimensional image feature vector f_{swin} extracted by the Swin Transformer, forming a composite feature vector f_{compose} . This composite vector is used for the final classification of ultrasonic standard planes. The entire process can be formulated as the following equation (7).

$$\begin{cases} t_i = \text{Linear}(\text{cls}, x, y, w, h, \text{conf}), & t_i \in \mathbb{R}^{64} \\ h_T = \text{GRU}(\{t_1, t_2, \dots, t_T\}), & h_T \in \mathbb{R}^{256} \\ f_{\text{compose}} = \text{Concat}(h_T, f_{\text{swin}}), & f_{\text{compose}} \in \mathbb{R}^{1024} \end{cases} \quad (7)$$

3.5. Plane Loss

Inspired by the research of Zeng et al. [22] on thyroid ultrasound standard plane images, FCNSU-NET effectively leverages the medical

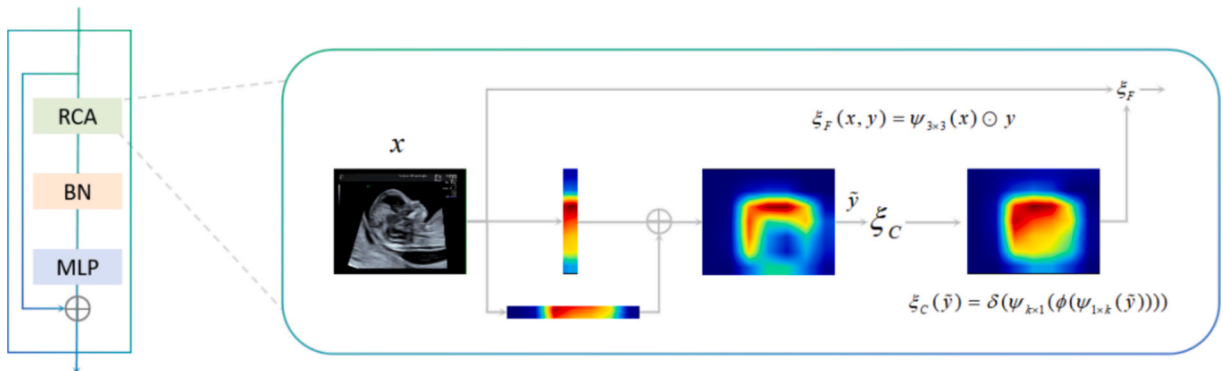


Fig. 6. RCM module structure diagram.

prior knowledge of FCNS-USP in the first trimester during the target detection stage, thereby enhancing the model's detection performance. For this task, we designed a tangent loss function, Plane Loss (P Loss), specifically for FCNS-USP in the first trimester. P Loss has two primary objectives: minimizing the number of incorrect plane targets detected and reducing the confidence of erroneously detected targets. To achieve these goals, targets detected by YOLO are first filtered, retaining only those with a confidence greater than 0.1, denoted as D_p . These targets are then validated against the expected anatomical set S_p for each plane, ensuring that they conform to the medical prior knowledge of the plane. The calculation formula of P Loss is shown in the following equation:

$$L_{plane} = \alpha \cdot \sqrt{N_{wrong}} + \beta \cdot \sum_{i \in D_p} (1 - I_{\{d_i \in S_p\}}) \cdot conf_i \quad (8)$$

Among them, $\alpha = 0.7$ and $\beta = 0.3$ are used to control the influence of each loss item on the overall loss; N_{wrong} represents the number of non-current plane targets detected. $D_p = \{d_i \in D | conf_i > t\}$ represents the set of detection targets whose confidence is greater than the threshold. $conf_i$ is the target's confidence level. $I_{\{d_i \in S_p\}}$ is an indicator function with a value of 1 if target d_i belongs to the set S_p of the expected anatomy of the current plane, and 0 otherwise. With root-mean-square error losses, we want to minimize the number of false checks so that the model has as little error detection as possible. The first part of the loss function is the quantity minimization loss, which minimizes the number of non-current plane targets detected by taking the square root. This allows the model to be punished more mildly when the number of false detection targets is small, avoiding oversensitivity to a small number of false detections. When the number of false detection targets is large, the growth rate of loss increases to reduce the occurrence of false detection. The second part of the loss function is the confidence reduction loss, by imposing a penalty on the confidence of non-current plane targets, the model will try to reduce the confidence of these error detections during training, thereby reducing the negative impact of these targets on the final model performance. When training the model, we update the weights of the model by adding our losses directly to our baseline model losses.

4. Experiments and results

4.1. Experimental setup

This study was approved by the Medical Ethics Committee ([2024] No. 394, Second Affiliated Hospital of Fujian Medical University), and the data were obtained from the hospital's ultrasound workstation. A total of 3,425 first-trimester fetal CNS ultrasound images were collected, including 3,052 images for model development and 373 images for clinical validation. The model development data were randomly split, with 80 % used for training and 20 % for testing. Clinical validation data were independently collected to evaluate the clinical effectiveness of the model (see Table 1 for dataset distribution).

All data were acquired by five physicians using 11 color Doppler ultrasound diagnostic instruments (convex array probe, frequency 7.5–12.0 MHz), including Mindray Resona 7, Resona 8, Resona I9S, GE Voluson E8, and Voluson E10, during first-trimester fetal ultrasound examinations. This ensured high reliability and diversity of the experimental data.

The experiments in this study were conducted on an Ubuntu 22.04 system with the following hardware configuration: Intel(R) Core(TM)

Table 1

Dataset distribution of fetal central nervous system ultrasound screening plane in the first trimester.

Data set	T-LV	C-FL	AP-MSP	T-Th	Total
Training set	764	1092	363	221	2440
Test set	192	273	91	56	612
Clinical validation set	99	90	100	84	373

i7-14700KF 3.40 GHz CPU, 64 GB RAM, and NVIDIA GTX-3090 GPU (24 GB VRAM). The programming environment used Python 3.10, with PyTorch 2.3.1 as the deep learning framework. Data analysis was performed using SPSS 20.0 and R 4.1.2. Continuous data were expressed as mean $\pm$ standard deviation ($\bar{x} \pm s$) and analyzed using a t -test; categorical data were expressed as frequencies or percentages and analyzed using the χ^2 test. The consistency between the AI model and the physicians' identification was evaluated using the Kappa test. Receiver operating characteristic (ROC) curves were plotted to assess the performance of primary, intermediate, and SPs, as well as the AI model, in recognizing standard planes and key anatomical structures, with the identification by the expert ultrasound team as the reference. The area under the curve (AUC) was compared using the DeLong test. A p -value of < 0.05 was considered statistically significant.

In this study, different evaluation metrics were used to assess the performance of the model in the tasks of plane recognition and anatomical structure detection. For the standard plane recognition task, Precision, Recall, Specificity, Accuracy, Kappa, and the Area Under the Curve (AUC) were used. Precision and Recall were used to evaluate the model's accuracy and coverage of positive samples, respectively, while Specificity was used to evaluate the model's ability to exclude non-target planes. Accuracy reflected the overall recognition correctness. Kappa was used to measure the agreement between the model's predictions and the judgments of the ultrasound specialist team, while AUC indicated the model's ability to distinguish between positive and negative samples.

In the anatomical structure detection tasks, Precision, Recall, and Mean Average Precision (mAP) are mainly used for evaluation. Precision and Recall measure the detection accuracy and coverage of the model, respectively. mAP@0.5 and mAP@0.5:0.95 are used to evaluate the model's detection performance under different Intersection over Union (IoU) thresholds. Among them, mAP@0.5 is a relatively lenient metric, while mAP@0.5:0.95 reflects the model's overall performance under more stringent conditions.

4.2. The detection results of anatomical structure in the first trimester FCNS-USP

Anatomical structure detection is a core task in this study. To validate the effectiveness of our improved model, we compared FCNSU-NET with several state-of-the-art models [37–46]. The results are shown in Table 2. In the anatomical structure detection task for first-trimester FCNS-USP, FCNSU-NET consistently outperformed all comparison models across key metrics.

Specifically, FCNSU-NET achieved a Precision of 95.0 %, an improvement of 5.6 percentage points over the baseline YOLOv8-m

Table 2

Performance of different models in detecting key anatomical structures in FCNS-USP.

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv5-m	86.4	87.1	89.3	0.552
YOLOv6-m	87.7	86.8	90.5	0.564
YOLOv7	88.1	87.4	90.6	0.561
YOLOv7-x	87.9	87.0	90.8	0.567
YOLOv8-m	89.4	89.1	91.8	0.589
YOLOv9-m	88.6	87.4	91.0	0.589
YOLOv10-m	86.9	87.8	91.2	0.568
YOLOv11-m	88.9	88.1	91.1	0.569
RT-DETR-L	86.5	87.1	90.6	0.558
Fast R-CNN (R-50)	—	—	89.5	0.537
DETR	—	—	87.8	0.513
CenterNet	—	—	90.4	0.538
EfficientDet	—	—	90.2	0.549
RetinaNet	—	—	88.5	0.523
FCNSU-NET (ours)	95.0(+5.6 %)	93.6(+4.5 %)	0.953(+3.5 %)	0.671(+8.2 %)

(89.4 %). Recall reached 93.6 %, 4.5 percentage points higher than YOLOv8-m (89.1 %), demonstrating better stability and completeness in capturing target anatomical structures and reducing false negatives. In terms of mAP, FCNSU-NET achieved an mAP@0.5 of 0.953 (3.5 % higher than baseline), indicating improved localization and identification accuracy under various threshold settings. For the more stringent mAP@0.5:0.95, FCNSU-NET achieved 0.671, which is significantly higher than other models. In particular, the model outperformed YOLOv9-m and YOLOv10-m by 8.2 percentage points, demonstrating more robust detection performance across a range of IoU thresholds.

To provide a more intuitive and detailed evaluation of detection performance, Fig. 8 visualizes the anatomical structure detection results from various models, with colored arrows indicating specific error types: red arrows highlight missed detections, orange arrows indicate inaccurate localization, purple arrows mark false positives, and yellow arrows denote multiple boxes for the same target.

These visualizations highlight the main detection challenges of different models. Missed detections (red arrows) often arise from targets with significant morphological variation or indistinct features, while multiple detections (yellow arrows) usually occur when boundaries are unclear or neighboring regions show similar patterns—often due to

artifacts. Inaccurate localization (orange arrows) is typically linked to blurred edges or complex shapes, and false positives (purple arrows) appear in regions where artifacts or background noise resemble real structures.

Notably, FCNSU-NET achieves more precise and consistent detection of targets within each standard plane, demonstrating superior performance in identifying key anatomical structures in every FCNS-USP standard plane. With the introduction of Plane Loss, Linear Transformer Attention, and CGRFPN, FCNSU-NET markedly reduces these errors. Plane Loss leverages prior knowledge to lower the rates of missed and false detections by informing the model of expected targets in each plane. Linear Transformer Attention strengthens global feature extraction, helping the model distinguish structures from complex backgrounds, while CGRFPN enhances multi-scale detection capability. These advances result in more accurate and reliable anatomical structure detection.

4.3. The recognition results of FCNS-USP plane in the first trimester

In addition to recognizing anatomical structures, FCNSU-NET also enables the recognition of FCNS-USP in the first trimester. For the task of

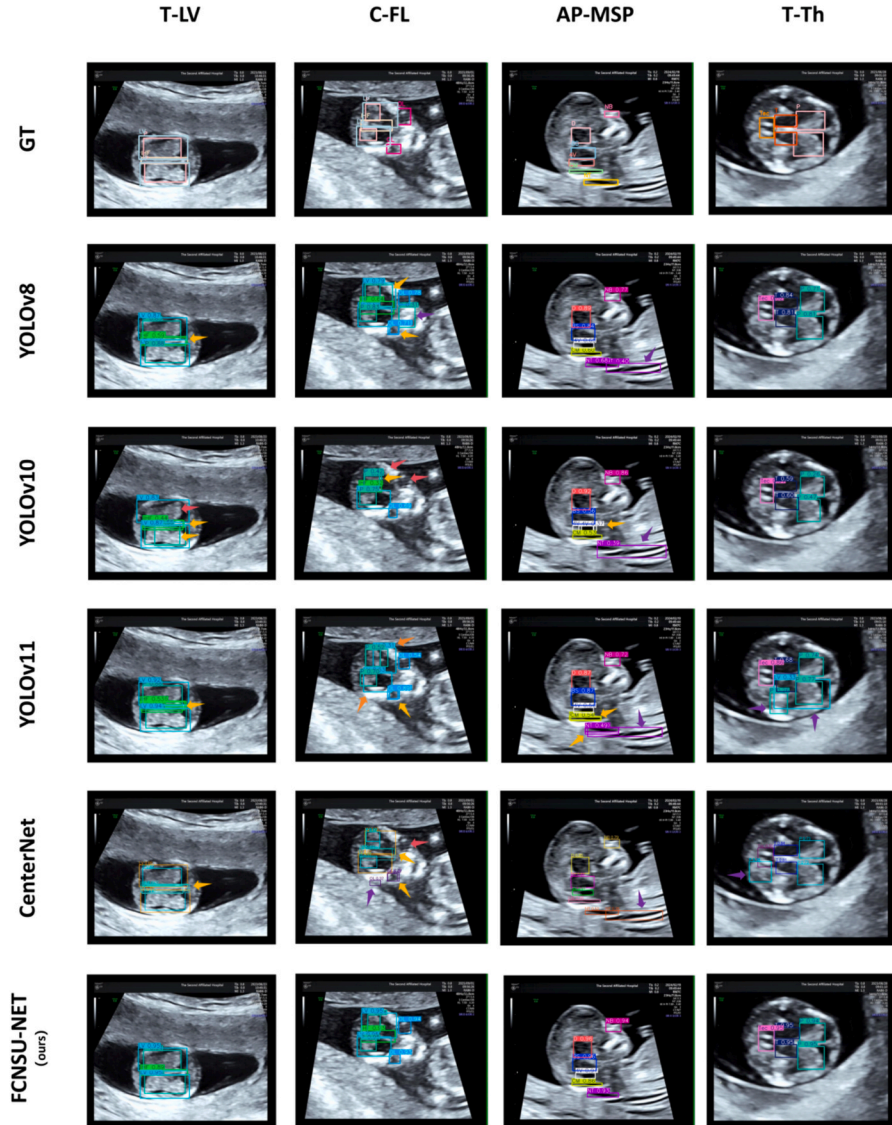


Fig. 8. Visualization of anatomical structure detection results in FCNS by different models.

plane recognition, we also compared the model with those that have better performance. The experimental results are shown in Table 3 below. In the FCNS-USP recognition task for the first trimester, it can be observed that the performance of FCNSU-NET significantly improved when anatomical structure information of the plane was added. Specifically, FCNSU-NET achieved an accuracy of 97.0 %, which is 9.3 percentage points higher than the best-performing Transformer model (85.6 % for Swin-T). In terms of recall rate, FCNSU-NET reached 98.9 %, a significant improvement over all other models, and 14.5 percentage points higher than the best-performing CNN model, ConvNeXt-T (84.2 %). This indicates that FCNSU-NET has significant advantages in correctly recognizing ultrasonic standard planes and reducing missed detection. Regarding specificity and accuracy, FCNSU-NET also performed well, reaching 99.6 % specificity, which was 3.0 percentage points higher than the second-best model (96.6 % for ConvNeXt-T), demonstrating its effectiveness in excluding false positives. The final accuracy was 98.53 %, an improvement of 8.5 percentage points over the best-performing other model (90.03 % for Swin-T), reflecting FCNSU-NET's strong overall capability in the standard plane recognition task.

4.4. Clinical validation of experimental results

To assess the clinical effectiveness and performance of the model, we conducted validation tests using the independent clinical validation dataset. Junior physicians (JPs), intermediate physicians (IPs), senior physicians (SPs), and the AI model independently interpreted the images in the validation set. The recognition results from each group were compared with those of the ultrasound expert team to evaluate consistency in standard plane recognition and the identification of key anatomical structures. Specifically, JPs were physicians who had completed standardized residency training, IPs had received over six months of specialized training in first-trimester fetal ultrasound, and SPs held prenatal diagnosis certificates with more than 15 years of prenatal ultrasound experience. All groups performed blind and independent identification, and the results were analyzed for agreement with expert team judgments. Performance comparisons for standard planes and key anatomical structures are summarized in Table 4 and Table 5, respectively.

Table 3
Performance of different models for FCNS-USP's four standard plane recognition.

Model	Precision	Recall	Specificity	Accuracy	Model
CNN	VGG[47]	80.6	71.0	93.7	83.3
	Resnet-50	82.7	81.0	95.9	88.07
	Resnet-101	85.0	78.6	95.2	86.4
	DenseNet-121[49]	79.4	74.8	94.6	84.5
	EfficientNet-v2[50]	85.4	77.3	95.1	86.92
	Mobilenet-v3	77.7	71.9	94.1	83.01
	ShuffleNet-v2	85.5	78.4	95.4	87.58
	ConvNeXt-T	85.6	84.4	96.6	89.86
	Swin-T[19]	87.7	84.2	96.4	90.03
	Swin-S[19]	85.1	82.4	96.0	88.56
Transformer	Vit-B/16[54]	79.2	79.3	95.0	85.13
	CSwin-T[55]	83.6	84.2	96.2	88.39
	CSwin-S[55]	86.6	79.8	95.5	87.91
	Agent Swin-T	85.2	84.2	96.4	89.54
	FCNSU-NET	97.0 (+9.3 %)	98.9 (+14.5 %)	99.6 (+3.0 %)	98.53 (+8.5 %)

Table 4

Recognition results of doctors and FCNSU-NET on four standard planes at different levels on the clinical data set.

Index	JP	IP	SP	FCNSU-NET
Sensitivity	60.35	76.20	83.48	96.10
Specificity	99.93	99.83	99.85	96.90
Positive predictive value	98.60	98.66	99.13	88.63
Negative predictive value	92.15	95.08	96.83	99.20
Accuracy rate	92.85	95.58	97.18	96.85
Kappa value	70.05	83.03	89.48	90.00
AUC	80.50	88.03	92.25	96.50

Analysis of the clinical validation data revealed that the AI model (FCNSU-NET) outperformed JPs and IPs in overall recognition, and achieved results comparable to or exceeding those of SPs in several metrics. For standard plane recognition, the sensitivity of the AI model was 96.10 %, substantially higher than that of JPs (60.35 %) and IPs (76.20 %), and slightly higher than that of SPs (83.48 %). While the specificity and negative predictive value of the AI model were slightly lower than those of human experts, its overall accuracy (96.85 %) was similar to that of SPs (97.18 %). The Kappa value reached 0.90, indicating a high level of agreement between the AI model and the expert team in standard plane recognition.

For key anatomical structure identification, the recall and accuracy of the AI model were comparable to or exceeded those of SPs for most structures. For example, the AI model demonstrated strong accuracy in recognizing the "brain lines" and "bilateral choroid plexus," underscoring its robust stability. However, its accuracy for structures such as the "nasal bone" and "bilateral orbit" was slightly lower than that of human experts, possibly due to the complexity of these anatomical features.

These findings indicate that FCNSU-NET holds considerable promise as an auxiliary tool for first-trimester fetal CNS ultrasound analysis, particularly in the sensitive identification of key anatomical structures. Further optimization of the model is expected to enhance its performance and further reduce the risk of misdiagnosis.

5. Discussion

5.1. Ablation studies

To systematically evaluate the contribution of each module to FCNSU-NET, we performed an ablation study by progressively adding GAT Block, CGRFPN, and Plane Loss (P loss), and observed the changes in model accuracy and efficiency. The results are presented in Table 6, which reports not only detection metrics (mAP@0.5, mAP@0.5:0.95), but also model computational complexity (GFLOPs) and average inference time per image (Time, ms), measured with batch size set to 1 and CUDA acceleration on the test set.

The introduction of the GAT Block led to improvements in mAP@0.5 (from 91.8 to 92.7) and mAP@0.5:0.95 (from 0.589 to 0.601), with a significant reduction in GFLOPs (from 79.1 to 50.6) due to the Agent Attention mechanism. This module reduces traditional self-attention complexity from $O(N^2)$ to $O(Nn)$, making model training and inference more efficient and suitable for large-scale datasets.

Adding the CGRFPN module further increased accuracy (mAP@0.5 to 94.3, mAP@0.5:0.95 to 0.638), with a slight increase in GFLOPs and a notable decrease in inference time (to 30.6 ms), demonstrating its effectiveness for multi-scale feature fusion while maintaining efficiency.

Finally, integrating the Plane Loss function further boosted detection performance (mAP@0.5 to 95.3, mAP@0.5:0.95 to 0.671), while keeping computational cost and inference speed steady (GFLOPs 56.7, Time 30.3 ms), highlighting the value of medical prior knowledge in enhancing consistency and accuracy.

Although Agent Attention greatly reduces theoretical computational complexity, it results in a moderate increase in inference time, likely due

Table 5

Recognition of anatomical results in four standard planes by physicians and FCNSU-NET at different levels on the clinical data set.

Plane	Structure	Recall				Precision				F1 score			
		JP	IP	SP	Ours Model	JP	IP	SP	Ours Model	JP	IP	SP	Ours Model
T-LV	IHF	92.6	96.5	96.8	92.8	95.6	91.2	98.9	98.9	94.1	93.8	97.8	95.7
	LV	90.8	91.0	94.6	94.9	96.7	98.9	95.7	100.0	93.7	94.8	95.1	97.4
	P	93.9	92.2	97.9	95.9	98.9	100.0	97.9	98.9	96.4	95.9	97.9	97.4
C-FL	NB	100.0	100.0	100.0	92.9	86.1	92.4	86.1	100.0	92.5	96.1	92.5	96.3
	D	96.4	97.1	93.9	96.4	98.8	82.7	95.1	98.8	97.6	89.3	94.5	97.6
	BS	95.9	100.0	98.6	94.8	94.6	97.3	96.0	98.7	95.2	98.6	97.3	96.7
	4 V	97.2	97.1	98.6	93.3	95.8	93.1	95.8	97.2	96.5	95.0	97.2	95.2
	CM	92.2	93.2	94.6	91.0	97.3	94.5	95.9	97.3	94.7	93.9	95.2	94.0
	NT	98.7	98.8	100.0	97.5	87.1	94.1	92.9	90.6	92.5	96.4	96.3	93.9
AP-MSP	Tec	100.0	100.0	98.8	93.7	84.3	83.2	91.0	100.0	91.5	90.8	94.7	96.7
	T	100.0	95.5	97.5	89.9	82.6	98.8	91.9	93.0	90.5	97.1	94.6	91.4
	P	100.0	96.3	94.5	95.2	68.9	85.6	95.6	88.9	81.6	90.6	95.0	92.0
T-Th	IHF	98.3	100.0	100.0	94.5	75.3	70.1	83.1	89.6	85.3	82.4	90.8	92.0
	OL	95.7	100.0	97.9	96.3	31.0	66.2	66.2	73.2	46.8	79.7	79.0	83.2
	LV	97.1	98.4	98.2	98.6	83.5	78.5	69.6	88.6	89.8	87.3	81.5	93.3
	P	98.3	98.4	98.4	97.0	84.3	85.7	90.0	92.9	90.8	91.6	94.0	94.9

Table 6

Results of ablation experiments on the influence of each module on FCNSU-NET detection performance.

GAT Block	CGRFPN	P loss	mAP@0.5	mAP@0.5:0.95	GFLOPs	Time (ms)
Baseline			91.8	0.589	79.1	8.6
✓			92.7	0.601	50.6	37.2
✓	✓		94.3	0.638	56.7	30.6
✓	✓	✓	95.3	0.671	56.7	30.3

to current CUDA parallelism and hardware utilization. Nevertheless, the actual inference speed remains much faster than that of the original Transformer and exceeds 25 fps, fully meeting real-time and clinical application requirements. These results underscore FCNSU-NET's potential as an efficient and reliable tool for clinical deployment, with further engineering optimization expected to yield even greater benefits.

5.2. Visualization of Heatmap for plane recognition

To evaluate whether the model can effectively focus on key anatomical structures in first-trimester FCNS-USP images, we employed Grad-CAM [57] to visualize the heatmaps of four standard planes, as shown in Fig. 9. For comprehensive comparison, we also visualized the heatmaps generated by our baseline image recognition model (Swin-T) and another high-performing model (ConvNeXT-T).

The comparative heatmaps show that both ConvNeXT-T and Swin-T are able to attend to some key anatomical structures in most planes, indicating that these models can extract part of the critical information needed for plane recognition. However, both models also tend to focus on irrelevant regions, such as the black background areas of the ultrasound images or non-essential image content, rather than being exclusively focused on diagnostically important structures. This dispersion of attention suggests that their feature extraction is less selective, potentially reducing their ability to consistently identify subtle anatomical cues.

In contrast, FCNSU-NET's heatmaps reveal markedly concentrated and accurate attention on the key anatomical structures, particularly in the T-LV and AP-MSP planes. For instance, the model consistently highlights regions corresponding to the bilateral lateral ventricles and brain midline in T-LV, as well as the nasal bone and diencephalon in AP-MSP. This focused response indicates that FCNSU-NET can more precisely capture and utilize relevant anatomical cues.

Such focused attention is closely associated with improved recognition performance. By encoding detected anatomical structures with GRU, FCNSU-NET can identify which targets are present in each plane

and focus on their corresponding anatomical regions. This allows the model to comprehensively consider both structure information and image features, placing greater emphasis on clinically important areas. This capability is particularly valuable in first-trimester scans, where anatomical structures are small and boundaries are unclear. As a result, FCNSU-NET achieves significantly higher accuracy and robustness in both standard plane recognition and anatomical structure detection tasks.

5.3. Limitations and future work

Despite the promising results demonstrated by FCNSU-NET, several limitations remain. First, the dataset used in this study was collected from a single medical center, which may limit the generalizability of the model to images acquired from other institutions or different ultrasound machines. Further multi-center validation and the inclusion of more diverse data are needed to ensure broader applicability.

Second, while the model performs well on standard plane and anatomical structure recognition tasks, its robustness against poor image quality, significant anatomical variation, and rare malformations requires further investigation. Challenging clinical cases such as severe artifacts or atypical fetal positions may still impact model performance.

Third, the current model focuses on the recognition of standard planes and key anatomical structures. Future work could extend FCNSU-NET to more complex clinical tasks, such as automated diagnosis of specific CNS abnormalities or integration with other imaging modalities for comprehensive prenatal screening. Moreover, although direct comparison with Transformer-based methods was performed (DETR and RT-DETR), future work will further explore comparisons with other emerging Transformer-based detectors tailored for medical imaging tasks to provide a more comprehensive evaluation.

Fourth, although the current workflow employs a modular design, future research will focus on developing a fully end-to-end model that directly performs standard plane and anatomical structure recognition from raw ultrasound images. This approach is expected to improve real-time performance and reduce error accumulation caused by multi-stage processing.

Finally, further optimization of hardware utilization, such as model quantization or operator-level optimization, will be explored to accelerate model inference and facilitate seamless deployment on edge devices in clinical workflows.

In future work, we plan to collect multi-center datasets, improve the model's robustness to complex clinical scenarios, develop end-to-end modeling approaches, and further enhance the performance of FCNSU-NET.

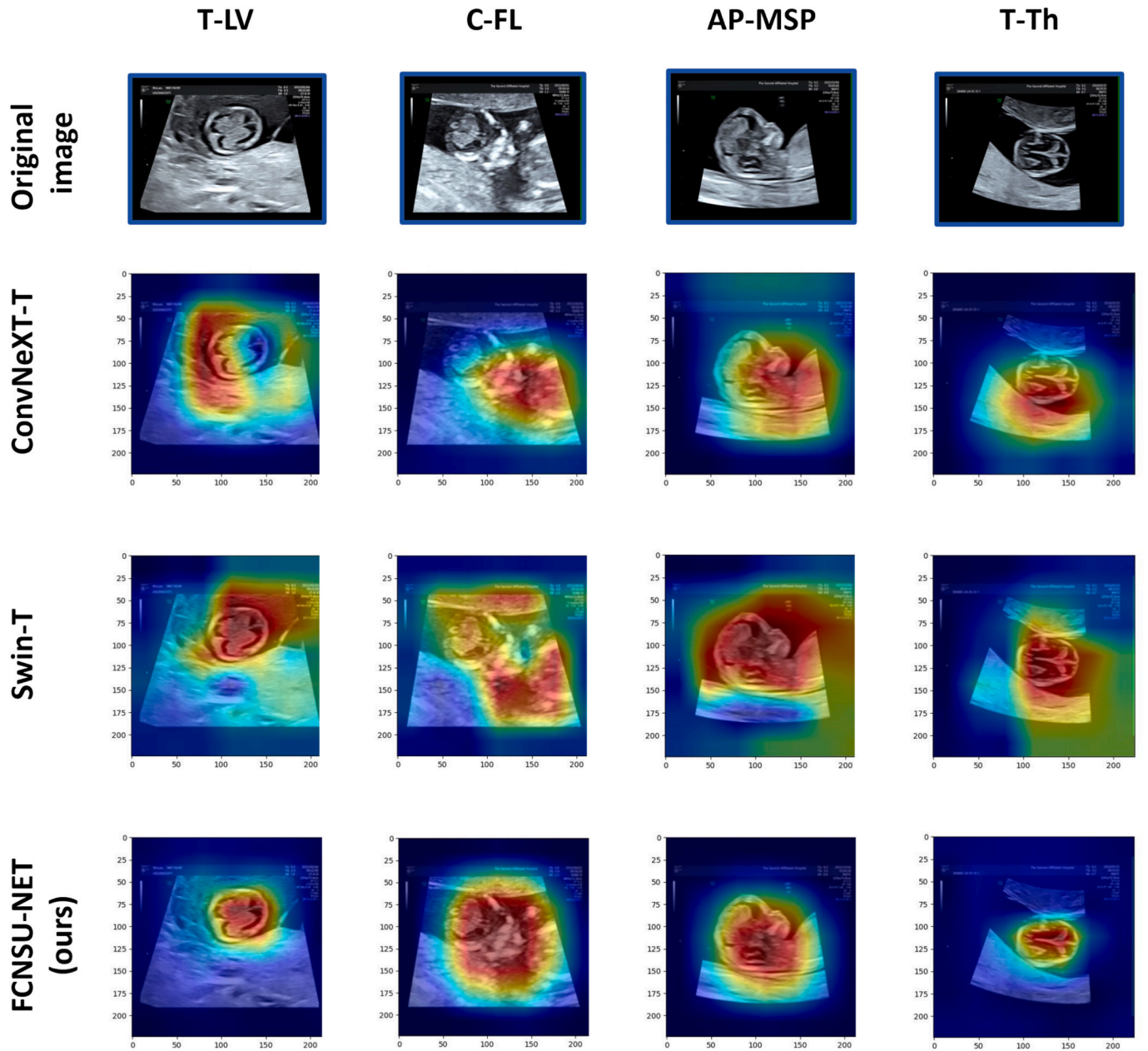


Fig. 9. Heatmap visualization when different models identify the four facets of FCNS-USP.

6. Conclusion

In this study, we proposed FCNSU-NET, an innovative deep learning model designed for comprehensive recognition of first-trimester fetal CNS ultrasound standard planes and key anatomical structures. By introducing modules such as Agent Attention, CGRFPN, Plane Loss, and GRU-based feature encoding, FCNSU-NET achieves significant improvements in both anatomical structure detection and plane recognition.

Specifically, the incorporation of Linear Transformer Attention not only reduces the overall parameter count and computational complexity, but also equips the model with powerful global attention capabilities, enabling it to effectively capture and utilize global image information. As a result, FCNSU-NET achieves high detection accuracy ($\text{mAP}@0.5 = 95.3\%$, $\text{mAP}@0.5:0.95 = 0.671$) and robust performance in standard plane classification (accuracy = 98.53 %, specificity = 99.6 %).

Clinical evaluation further demonstrates that FCNSU-NET's

diagnostic performance exceeds that of less-experienced clinicians and matches or even surpasses senior experts ($\text{Kappa} = 0.90$), validating the robustness and generalizability of the model in real-world clinical scenarios.

Importantly, by introducing Linear Transformer Attention, the model achieves lower parameter count, enhanced global feature modeling, and real-time inference (over 25 fps), ensuring both efficiency and accuracy for clinical application. Future work will continue to improve performance and clinical applicability.

CRediT authorship contribution statement

Qingyu Zheng: Writing – original draft, Methodology, Investigation, Data curation. **Liya Li:** Data curation. **Yubing Huang:** Project administration, Conceptualization. **Guorong Lyu:** Writing – review & editing. **Pan Zeng:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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